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EWPTNN: An Efficient Workload Prediction Model in Cloud Computing Using Two-Stage Neural Networks

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Abstract

Allocating insufficient resources to the cloud applications, leading to the loss of revenues, consumers and, Quality-of-Services (QoS). Another side, allocating resources more than is needed leading to the wastage of energy and cost to maintain the resources like servers, datacenters and, network bandwidth etc. So, this problem can be solved with predictive scaling methods by using machine learning approaches, which can estimate the future needed resources and, performing scaling operations in advance. To perform accurate scaling operations, authors presented an enhanced prediction model based on a neural networks, which is a combination of a classification and prediction model to predict the accurate utilization of resources. At the first stage, the proposed model categorizes the resources into three classes like over, normal and, under adaptively based on the utilization of resources. In the next stage, it predicts the future utilization of resources by using neural network regression model based on the classification results. The experimental results showed that the proposed model achieved accurate results.

Keywords: Artificial Neural Networks; Cloud Computing; Classification; Prediction; Enterprise Workloads.

1. Introduction

In present days, cloud computing providing many services in computing environment while providing computing infrastructure to organizations and personal users without physical infrastructure. Providing technical processing capacity to analyze the data aimed to reach the goal of high performance computing. This idea enables small organizations also can access a high performance computing power without the maintenance of complex infrastructure

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like datacenters, servers and, bandwidth etc. With the help of cloud computing technology, start-up companies also providing many services while avoiding the issue of high power consumption [1]. The main aim of the cloud computing technology is to provide on-demand services with scalability, availability and, reliability in a distributed computing. Cloud computing providing mainly three types of services such as Software-as-a-Service (SaaS), Platform-as-a-Service (PaaS) and, Infrastructure-as-a-Service (IaaS). But nowadays cloud computing is treating as everything as a service (XaaS). Virtualization is one of the key features of cloud computing technology which refers to the partition of logical resources from physical infrastructure in order to reduce cost and enhance flexibility [2]. Dynamic load balancing is another main feature of the cloud computing technology which provides the self-regulating mechanism to handle the workloads [3]. This mechanism helps to allocation and reallocation the workload among resources to reach the maximum throughput while reducing the response time, power consumption and, cost. On the other side, improving the resource utilization and performance of the resources. The main aim of the cloud computing paradigm is to provide on-demand resources provisioning while satisfying economically to both the cloud providers and cloud users. Due to the maximum usage of the cloud computing paradigm in market-oriented, the traditional resource balancing management techniques need to be enhanced. The traditional resource management techniques are not sufficient to cloud providers and handle huge workloads.

In cloud computing, dynamic scaling of resources can be done in two ways such as reactive approach and proactive approach [4]. In reactive approaches, cloud provider fixes the threshold values for over-utilization state and under-utilization state. When workload reaches the threshold value, the provider takes the action based on the state of resources, like either adds virtual machines (VMs) or remove virtual machines from cloud services. In the case of sudden changes in workload, the provider faces difficulties to perform the scaling operations. The proactive approach is another way, to perform scaling operations of resources in advance. Cloud provider predicts the workload for each cloud service which is deployed in a SaaS platform and allocate the resources to cloud service based on the predicted value. Many prediction approaches are being used for predicting the workload in the cloud environment. Few kinds of prediction methods used for predicting time series models such as ARIMA model, Support vector machine, Neural network, Bayesian model and, Moving average model etc. Another type of prediction methods used for identifying patterns of the cloud workload. But to improve the accuracy for prediction approaches is a challenging task due to the various kinds of cloud services. To estimate the right amount of resources to the cloud services, proposed an efficient workload prediction model by using two stages of a neural network model. At the first stage, categorizing the VMs based on the utilization (CPU and memory) by using a classification model. In the second stage, predicting the future workload value for classified VMs. According to the prediction value, VM manager performs the scaling operations in advance. The major contributions of this paper are as follows:

- Proposed a two-stage neural network model which can be used for classification of VMs at the first stage and VM load prediction at the second stage.
- Considered two parameters of VM which are CPU utilization and memory utilization for identifying each VM load.
- Considered bit-brains cloud workload dataset to evaluate the proposed model.

The remainder part of the paper is organized as follows: The problem specifications are presented in Section 2. In section 3, related work is presented in Section 3. In Section 4, presented a proposed methodology with architecture. Section 5 explained about results and evaluations. Finally, conclude the paper in section 6.

2. Related work

Cloud resources workload can be represented as a time series. A time-series is an episode of points measured in time-space at unique time intervals. To predict the future value from time series, several authors have proposed different methods. Authors [5], [6] presents a survey on the prediction methods of cloud services in several aspects like workload and performance of the cloud service. Surveyed on different resource provisioning issues and resource provisioning algorithms in a cloud environment. Discussed several prediction methods in machine learning for cloud services and presented a taxonomy for prediction approaches. In [7], the authors developed a prediction approach to predict the cloud workload by using autoregression (AR) and moving average (MA) methods. Author [8] developed a prediction framework to find out the future value of the application workload. This framework is developed by using a combination of neural networks (NN) and linear regression (LR) methods. Author [9] proposed a prediction framework with the help of neural network methods. Considered a resource load for the prediction process. In [10], the authors proposed a hybrid prediction method to improve the accuracy in the prediction process. This framework is a combination of fuzzy neural networks and moving average methods. At the first stage, using moving average methods to predict workload fluctuations. In the next stage, using fuzzy neural network rules to improve the accuracy rate. Another author [11] proposed NN based method to predict the CPU utilization of VMs. In [12], the author presented an algorithm to perform the scaling operations in advance. This proactive scaling algorithm is based on fuzzy association rules. Additionally, the authors used genetic algorithms and back-propagation method to optimize the network. Considered SLA violation parameter to evaluate the results. Author [13], proposed an enhanced resource management system to improve the QoS by using fuzzy time series model. Author [14], developed an algorithm by using fuzzy logic and control theory methods to allocate resources in datacenters. This method predicts the workload of datacenter.

3. Problem specification

Cloud computing is an emerging technology which provides different kinds of services in heterogeneous environments such as Internet-of-Things, enterprise computing, and big-data analytics. But it is still facing several challenges and issues, like resources balancing. Such resources balancing for processing the continuous streaming data is challenging task due to (i) uncertain request arrival rate (ii) need to analyze different patterns from collected data (iii) uncertain consumptions of resources. To solve the issues mentioned above, proactive approaches can be solved effectively with the help of machine learning approaches. These machine learning methods using historical workload records of cloud applications which are available for a particular time period in the past. The main aim of the machine learning approaches is, creates intelligent resource management system based on the previous data. However, the existing prediction methods are not able to handle complex data structures and not making accurate estimations. If prediction algorithm gives the wrong estimation like the next state of VM is over-utilization for the next interval, then allocated extra resources will be wasted and, it can lead to maximizing the maintenance cost of infrastructure, high power consumption and, high Co_2 emission. On the other side, if prediction algorithm gives another wrong estimation like the next state of VM is under-utilization for the next interval, then cloud provider performs scale-in operations to reduce the number of VMs, it can lead to dropping the QoS and losing the revenues and consumers. The proposed method aims to enhance the resource management system by improving the accuracy of predictions. So that, VM manager can perform accurate scaling operations of cloud resources in advance. To handle the millions of VMs while improving the prediction accuracy, the proposed approach is a very useful methodology.

4. Proposed methodology

In this section, presented a proposed methodology to classify the VMs in a cloud datacenters based on the state of utilization at the first phase. Later, using a prediction algorithm estimating individual VM utilization. The main objective is to classify the VMs according to their utilization.

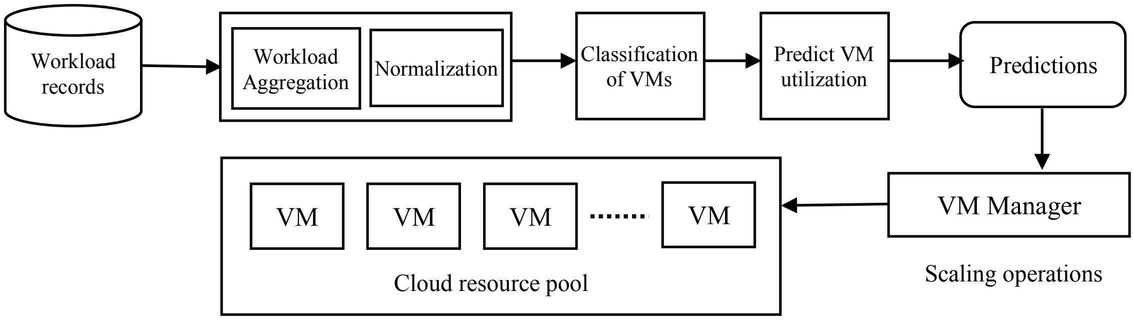

Fig. 1. Workflow of the EWPTNN model

The proposed method as shown in the Figure 1, consists of the following steps,

- Using a multilayer perceptron neural network, classifying the VMs into three categories based on the utilization.
- Using a neural network regression, predicting the VM workload.
- According to the prediction results, a VM manager performs the scaling operations in advance.

4.1 Preprocessing

Neural network layer consists of logical units which are called neurons. The proposed NN model is consists of a four-layered neural network. To utilize the multilayer neural network for the proposed approach, the VM utilization measured time-space is stored in historical workload records to use as inputs for the prediction model. The dataset contains utilization percentages of CPU and memory. The input data is considered as utilization of VM which is a combination of CPU utilization and memory utilization to train the network. In the preprocessing step, first performs the data aggregation method to convert the data into a unique time interval (60 min time interval). In the next step, by using min-max normalization process data obtained from a dataset. Here CPU utilization and memory utilization normalized in the range of (0,100) by using equation 1. The input data can be formed as shown in equation 2. After preprocessing the data, data is separated into two parts of datasets which are called as training dataset and testing dataset. In the complete dataset, 70% of the data is used for the training process and the remaining 30% of the data is used for evaluating the accuracy of the prediction model for two phases.

$$\overline{X_i} = \frac{X_i - X_{min}}{X_{max} - X_{min}} \quad (1)$$

$$\begin{bmatrix} 0.93 & 0.84 & 0.75 & 0.62 & 0.75 & 0.71 & \cdots \\ 0.84 & 0.75 & 0.62 & 0.75 & 0.71 & 0.69 & \cdots \\ 0.75 & 0.62 & 0.75 & 0.71 & 0.69 & 0.68 & \ldots \\ 0.62 & 0.75 & 0.71 & 0.69 & 0.68 & 0.57 & \cdots \\ 0.75 & 0.71 & 0.69 & 0.68 & 0.57 & 0.55 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \cdots \end{bmatrix} \quad (2)$$

4.2 Classification Approach

After the preprocessing step, normalized data can be used as input to the network. In this phase, by using a multilayer perceptron network classifying the VMs into three major categories based on the CPU utilization and memory utilization. These categories are (i) Over-utilization VMs (ii) Normal-utilization VMs (iii) Under-utilization VMs. Categorizing the VMs according to the CPU and memory utilization is presented as a time series, with utilization percentage in between 0 and 100. The proposed method considers only over-utilization VMs and under-utilization VMs for further processing. Proposed model main aims to get an accurate state about over-utilization and under-utilization VMs. As shown in Figure 2(a), the network model classifying the workload records into three categories as mentioned above.

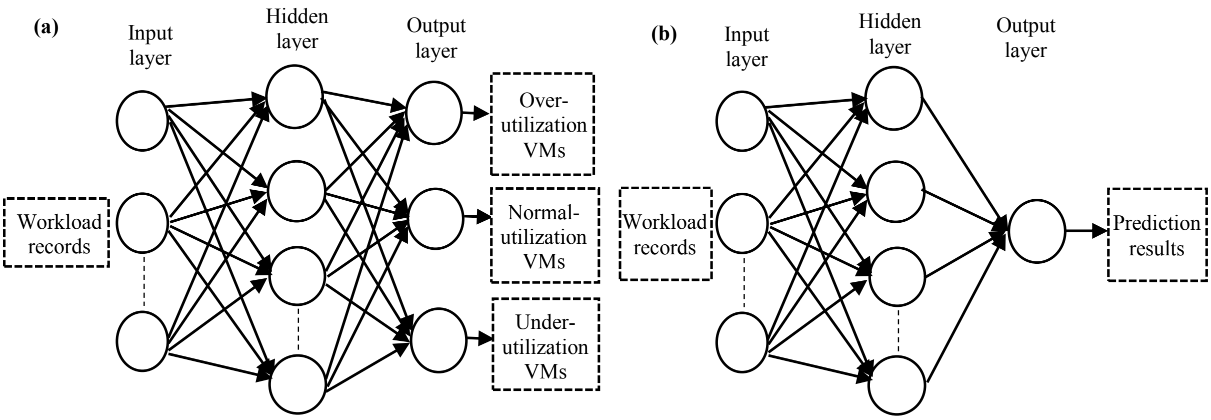

Fig. 2. (a) Classification of VMs (b) Workload prediction of VMs

4.3 Prediction Approach

In the second phase based on the classification neural network model results, NN regression model is trained to be with two categories of data. The second phase prediction model uses time-series data to predict the future utilization of VMs. As presented in Figure 2(b), the neural network regression model predicts the future utilization of over-utilized and under-utilized VMs. This model is similar to the classification prediction model, but this model consists of a single neuron in the output layer. This output neuron provides the continuous value of the time series. The second phase prediction model analyzes the previous VM utilization values to predict the future utilization of VM at time interval t+1.

Further, based on future utilization values, VM manager performs the scaling operations in advance. The quality scaling operations of resources represents the accuracy of the prediction model. Finally, the VM manager performs either horizontal scaling or vertical scaling based on results, such as increasing or decreasing shared resources and, VMs.

5. Results and Discussion

In this section, evaluated the results of the proposed approach which was presented in the previous section. Performed the tests on the training dataset and testing datasets. The Proposed model consists of two types of prediction approaches such as classification approach and prediction approach, so results section was categorized into two phases. In the first phase evaluated the classification model and, the prediction model was evaluated in the second phase.

5.1 Datasets

To evaluate the proposed approach, considered GWA-T-12 bitbrains (from distributed datacenter) dataset. This dataset contains two categories of storage information (SAN) about VMs such as fast-storage and Rnd. We considered fast-storage dataset to evaluate the proposed approach. The dataset contains the performance metrics of 1250 VMs. For each VM, data is stored in '.CSV' file which contains the performance metrics of individual VM. Each file having 11 metrics information, among those metrics considered CPU utilization and memory utilization metrics to train the network. Out of 1250 files, 900 files used for training purpose and, the remaining 350 files used for testing purpose.

5.2 Phase1 results

Phase1 consists classification model to classify the VMs into three categories. This model is categorizing the VMs into three classes based on the utilization of VM. This classification network used multilayer perceptron classifier to classify the VMs. To reduce the classification error, used an iterative method to update the weights of neurons. To evaluate any classification method, parameters can use such as precision, recall and, F1-measure. To calculate the precision, recall and, F1-measure for each class, plotted the confusion matrix based on the table1. In this table, for three classes given names based on the utilization such as over (O), normal (N) and under (U), and TP, E represented as true positive rate and error rate respectively according to the class. In confusion matrix, row values represent actual classes and column values are represents predicted classes. With the help of plotted confusion matrix, calculated all three parameter values as mentioned above. For example, the proposed classification algorithms achieved precision, recall and, F1-measure of 95.5%, 95.1% and, 95.3% respectively for under-utilization class.

Table 1. Confusion matrix for three classes.

		Predicted classes		
		Under	Normal	Over
Actual classes	Under	TP_U	E_{UN}	E_{UO}
	Normal	E_{NU}	TP_N	E_{NO}
	Over	E_{OU}	E_{ON}	TP_O

For individual class, we calculated the precision, recall and, F1-measure by using the following equations. If calculating all three metrics for under-utilization class then equations are:

$$Precision_{under} = \frac{TP_U}{TP_U+E_{NU}+E_{OU}} \quad (3)$$

$$Recall_{under} = \frac{TP_U}{TP_U+E_{UN}+E_{UO}} \quad (4)$$

$$F1 - measure_{under} = \frac{2*Precision*Recall}{Precision+Recall} \quad (5)$$

Table 2. Evaluation of three classes

	Precision	Recall	F1-measure
Under	0.955	0.951	0.953
Normal	0.943	0.921	0.931
Over	0.941	0.911	0.925

5.3 Phase2 results

Phase2 consists another prediction method to predict the future utilization of VM. The prediction network used neural network regression to predict future VM utilization. To train the network for prediction, considered preprocessed data. To evaluate the model, the last column of data from equation 2 is stored in 'Y' vector as shown in equation 6. Accuracy is evaluated by using mean squared error (MSE), as shown in the equation 7. In equation 7, $\overline{Y}_i$ and Y_i are the predicted and actual utilizations respectively at time interval 'i'. The error rate of the proposed model is 0.132 for the next 60 min time interval.

$$X = \begin{bmatrix} 0.93 & 0.84 & 0.75 & 0.62 & 0.75 \\ 0.84 & 0.75 & 0.62 & 0.75 & 0.71 \\ 0.75 & 0.62 & 0.75 & 0.71 & 0.69 \\ 0.62 & 0.75 & 0.71 & 0.69 & 0.68 \\ 0.75 & 0.71 & 0.69 & 0.68 & 0.57 \end{bmatrix} \quad Y = \begin{bmatrix} 0.71 \\ 0.69 \\ 0.68 \\ 0.57 \\ 0.55 \end{bmatrix} \quad (6)$$

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\overline{Y}_i - Y_i)^2 \quad (7)$$

Table 3. MSE of proposed approach

Prediction interval (min)	MSE error
05	0.003
10	0.052
20	0.086
30	0.102
50	0.119
60	0.132

With experimental results on the training dataset and testing dataset, shown that with the help of proposed approach cloud provider can enhance the resource management system while performing the scaling operations in advance.

6. Conclusion

The main aim of this paper is, to represent the enhanced approach for a cloud provider to perform the scaling operations in advance while maximizing the QoS by providing sufficient resources to cloud users in time and minimizing the power consumption and high CO_2 emission by utilizing the maximum amount of resources of the datacenter. The proposed approach used a multilayer perceptron classification method to categorize the VMs based on the utilization of VM. In next the stage considered over-utilized and under-utilized VMs to predict the future utilization of their VMs. Finally, the VM manager performs the scaling operations in advance based on the predicted results. The experimental results showed that the proposed model achieved accurate results.

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